

B6.2 Feeding the human race

Summary

The human population is increasing rapidly and with this comes a need for more food. Biologists are seeking to tackle this increased demand, which will lead to an improvement in the lives of many people around the world. However, there are many things to consider in achieving this aim, not least the impact on ecosystems. There is much debate surrounding the use of gene technology as a potential solution to the problem of food security.

Underlying knowledge and understanding

Learners should be familiar with the content of a healthy human diet and the consequences of imbalances in a healthy daily diet. Their knowledge and understanding from topics 1, 4 and 5 will also be drawn together in this topic.

This includes the organisation of DNA, what plants require enabling them to photosynthesise, interactions between species and the idea of variability within species and subsequent selection of characteristics.

Common misconceptions

Learners can often think that genetic engineering leads to the increased use of pesticides.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
BM6.2i	extract and interpret information from charts, graphs and tables	M2c and M4a

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
B6.2a ☑ describe some of the biological factors affecting levels of food security	increasing human population, changing diets in wealthier populations, new pests and pathogens, environmental change, sustainability and cost of agricultural inputs	M2b, M2f		
B6.2b ☑ describe and explain some possible agricultural solutions to the demands of the growing human population	increased use of hydroponics, biological control, gene technology, fertilisers and pesticides		WS1.1c	
B6.2c explain the impact of the selective breeding of food plants and domesticated animals		M1c, M2c, M4a	WS1.1c	Research into the Rothamsted Research Broadbalk experiment.
B6.2d describe genetic engineering as a process which involves modifying the genome of an organism to introduce desirable characteristics				
B6.2e describe the main steps in the process of genetic engineering	restriction enzymes, sticky ends, ligase, host bacteria and selection using antibiotic resistance markers, vectors e.g. plasmids			Produce a storyboard of the processes for genetic engineering.
B6.2f explain some of the possible benefits and risks of using gene technology in modern agriculture	practical and ethical considerations		WS1.1c, WS1.1d, WS1.1e, WS1.1f, WS1.1g, WS1.1h, WS1.1i	Research into the advantages and disadvantages of selective breeding and genetic engineering.
B6.2g ☑ describe and explain some possible biotechnological solutions to the demands of the growing human population	genetic modification	M1c, M2c, M4a	WS1.1c, WS1.1g	Research into the growth of GM crops or livestock.